



EAST WEST UNIVERSITY

Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering Program
Final Examination, Spring 2025 Semester

Course: CSE489 Mobile Programming Section-01
Instructor: Dr. Md Mostofa Kamal Rasel, Associate Professor.
Full Marks: 50 (25 will be counted)
Time: 1 Hour 20 Minutes
Notes: Answer all questions. The marks of each question are listed on the right.

1. *TicTacPlay* is a turn-based two-player Tic-Tac-Toe game designed for Android devices. The game can be played in both offline and online modes. In the online mode, once a player makes a move, the opponent must be notified either instantly or after a countdown finishes. To ensure fair gameplay, each player must make their move within a limited time during their turn. [CO2,C3 Marks: 8]
Suppose you have a *TextView* named *tvTimer* that displays the countdown timer for the current player's move. **Write** the Android Java code snippet to update the countdown every second using any Android-supported threading mechanism.
2. The *MyDownloadTask* class below is a basic implementation of an asynchronous task to download a video file from a remote source. Each method contains some information and hints about its purpose. However, some of these pieces of information or hints are incorrect. Your task is to **identify** the incorrect ones and **replace** them with the correct information or hints. [CO2,C2 Marks: 7]

The *MyDownloadTask* class below is a basic implementation of an asynchronous task to download a video file from a remote source. Each method contains some information and hints about its purpose. However, some of these pieces of information or hints are incorrect. Your task is to **identify** the incorrect ones and **replace** them with the correct information or hints.

```
public class MyDownloadTask {  
    // Info-1: Statements of this block are executed in a worker thread  
    // Hint-1: invoke http(s) request to start downloading here  
    new Thread(new Runnable() {  
        public void run() {  
            // Info-2: Statements of this block are executed in a worker thread  
            // Hint-2: Accumulate downloaded data here  
            byte[] buffer = loadPartOfData(); // statement-1  
            progress = computeProgress(buffer.length); // statement-2  
            // Info-3: while-loop is executed in UI thread  
            while (progress < 100) { // statement-3  
                threadHandler.post(new Runnable() {  
                    public void run() {  
                        progressBar.setProgress(progress); // statement-4  
                    }  
                });  
                Thread.sleep(1000); // statement-5  
            } // end-of-while  
  
            // Info-4: Statements of this block are executed in the UI thread  
            threadHandler.post(new Runnable() {  
                public void run() {  
                    // Hint-3: send a message from worker to UI thread from here  
                    status.setText("Completed !!"); // statement-6  
                }  
            });  
        } // end-of-run()  
    }).start();  
}
```

3. Suppose that you are developing an Android app called RecipeMaster, which helps users explore and save cooking recipes. The app has two screens: HomeActivity, which displays a list of popular recipes, and DetailActivity, which shows full instructions, ingredients, and images for a selected recipe. In HomeActivity, users can tap a "Save" button on a recipe to download it for offline viewing, or tap the item to view more details. In DetailActivity, users can also save the recipe. The app must allow only one recipe download at a time, and downloads cannot be cancelled, even if the app is closed. Additionally, saved recipes should persist and be reloaded automatically even after the device is restarted. The download or save progress must be visible and updated on both screens, regardless of which one the user is currently using. [CO3,C3
Marks: 15]
- Now, answer the following questions:
- a) **Prepare** a list of required features for the app. (2 marks)
 - b) **Select** an appropriate Android element or component for implementing each feature. (4 marks)
 - c) **Implement** Java code snippet (only that is essential) to accomplish these features. (9 marks)
4. *Lost & Found* is a mobile application designed to help users share information about found items and search for lost ones. The app must store various types of data, including both public and sensitive information. For example, user location and contact numbers are considered highly sensitive and must be stored securely. Access to such data should be restricted—only verified and confirmed users who are directly involved with a specific lost or found item should be able to view another user's sensitive details. [CO3,C3
Marks: 8]
- a) Considering that all types of data must be stored and accessed remotely and securely, **suggest** your preferred approach—Own Processing Technique, API, or WebView—for handling the following aspects: implementation, authentication and verification, performance optimization, and data security. (2 marks)
 - b) **Implement** the XML and Java code necessary to load and display the website <http://www.lostnfound.info> using a WebView component. (6 marks)
5. Assume you have developed the *QuizBank* mobile application, which allows users to practice and take quizzes on various academic and competitive subjects. The app provides question banks, timed tests, performance tracking, and subject-wise progress reports. Now, you aim to generate revenue from it. **Propose** a suitable business model—either individual or hybrid—that ensures continuous income from users while they are actively using the app. Clearly **explain** how the chosen model enables consistent monetization from users over time. [CO3,C2
Marks: 7]
6. Suppose that you have finished developing a mobile application and are ready to publish it on the Google Play Store. Now, **identify** the general requirements for publishing an Android app on the Play Store, including necessary account types, required assets, and compliance with content policies. [CO3,C2
Marks: 5]